or

$$\begin{bmatrix} q_{n+1} \\ p_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & h \\ -h & 1 \end{bmatrix} \begin{bmatrix} q_n \\ p_n \end{bmatrix}$$

and we see that the eigenvalues of the propagation matrix are outside the unit disk. Thus both solution components tend to infinity unless $(q_0, p_0) = (0, 0)$. This means that with the single exception $\rho(q, p, 0) = \delta[q]\delta[p]$ there is no limiting (invariant) distribution for Forward Euler. To put it another way, the only solution, in the weak sense, of $\mathcal{L}_{\tilde{f}}^{\dagger}\rho = 0$ is the point distribution $\delta[q]\delta[p]$, and this solution is not "asymptotically stable."

As an alternative, consider the Backward Euler method: $q_{n+1} = q_n + hp_{n+1}$, $p_{n+1} = p_n - hq_{n+1}$. Then it is easily shown that both eigenvalues of the matrix lie in the interior of the unit disk, and hence all solutions of the recurrence relation tend to the origin with increasing n (the origin is an attractive equilibrium point). In this case, all densities evolve toward the Dirac distribution centered at the origin: $\delta[q]\delta[p]$. The only (distributional) solution of $\mathcal{L}_{\tilde{f}}\rho = 0$ is again $\delta[q]\delta[p]$, which in this case is attractive.

Now consider the application of the Symplectic Euler method, $q_{n+1} = q_n + hp_{n+1}$, $p_{n+1} = p_n - hq_n$. As we know, the solution through any given initial point remains confined to an ellipse for all n defined by $\tilde{H}_h(q, p) = \tilde{H}_h(q_0, p_0)$, where $\tilde{H}_h$ is the perturbed Hamiltonian defined by the backward error analysis. This implies that a density which is a function of $\tilde{H}_h$ will remain invariant under the associated Liouville equation. There are many solutions of $\mathcal{L}_{\tilde{f}}^{\dagger}\rho = 0$, none of which is attractive.

The harmonic oscillator is of course an integrable system. When the underlying system is a chaotic Hamiltonian system, the trajectories from any collection of initial conditions become disordered (even while preserving the energetic constraints defined by their initial states). The hope (in some cases, naive) is that typical trajectories computed using a symplectic method applied to such a chaotic system provide space filling curves on the $\tilde{H}_h = \mathrm{E}$ perturbed energy surface.

Example 5.7 (Ergodicity in the Trimer) At positive energies, the trimer is unstable and the atoms separate, into a bound pair and a single atom. At very low (but admissible) energies, the timer has an insurmountable energy barrier between the two basins. At moderate energies, the trimer will visit both basins, however, the mean passage time between basins may be greatly extended depending on the energy level.

Consider the level set of potential corresponding to $U = -0.5$. Note that the kinetic energy is everywhere positive, so that a trajectory with total energy $\mathrm{E} = -0.5$ will have to be bounded within the $U = -0.5$ level set. Would the dynamics (started from an initial condition with total energy $\mathrm{E} = -0.5$) completely fill in this level set over time? Because this system is chaotic, we cannot write down a formula for the exact solution, or find exact trajectories of the system, but we can use numerical methods to attempt to shed some light on the question. In Fig. 5.8 we